

Customer Shopping Behaviour Analysis

Business Problem Statement

A leading retail company wants to gain a deeper understanding of its customers' shopping behaviour to improve **sales performance, customer satisfaction, and long-term customer loyalty**.

The company has observed variations in purchasing patterns across **customer demographics, product categories, shopping channels, discounts, customer reviews, seasons, and payment preferences**. Management wants to identify the key factors influencing customer purchasing decisions and understand the behaviour of repeat and loyal customers.

The objective of this project is to analyze the company's customer shopping data and answer the following key business question:

How can the company leverage customer shopping data to identify purchasing trends, improve customer engagement, and optimize marketing, sales, and product strategies?

Project Deliverables

1. Data Preparation & Modeling — Python

Clean, transform, and prepare the raw customer shopping dataset for further analysis.

This includes:

- Data cleaning and preprocessing.
- Handling missing and inconsistent data.
- Data type conversion.
- Feature transformation and creation.
- Exploratory Data Analysis (EDA).
- Preparing the final dataset for SQL analysis and Power BI visualization.

2. Data Analysis — SQL

Load the prepared dataset into a relational database and perform business-focused analysis using SQL.

The analysis will focus on:

- Customer segmentation.
- Purchasing behaviour and revenue.
- Product and category performance.
- Discount effectiveness.
- Customer loyalty and repeat purchases.
- Subscription behaviour.
- Payment and shipping preferences.

- Identifying key purchasing patterns and drivers.

Advanced SQL concepts such as **subqueries, CTEs, CASE statements, aggregations, and window functions** will be used to generate meaningful business insights.

3. Visualization & Insights — Power BI

Develop an interactive Power BI dashboard to provide stakeholders with a clear view of customer shopping behaviour.

The dashboard will highlight:

- Sales and revenue KPIs.
- Customer demographics.
- Product and category performance.
- Discount usage and impact.
- Customer segmentation and loyalty.
- Subscription behaviour.
- Payment and shipping preferences.
- Purchasing trends and patterns.
- Interactive filters for deeper analysis.

4. Business Report & Presentation

Prepare a comprehensive project report covering:

- Business problem and objectives.
- Dataset overview.
- Data preparation methodology.
- SQL analysis.
- Power BI dashboard.
- Key findings and insights.
- Business recommendations.

A presentation will also be created to communicate the major findings, insights, and actionable recommendations to stakeholders in a clear and visually engaging manner.

5. GitHub Repository

Create a well-structured GitHub repository containing all project resources, including:

- Python scripts/notebooks.
- Cleaned and processed datasets.
- SQL queries.
- Power BI dashboard (.pbix).

- Project report.
- Presentation.
- Documentation and README file.

Overall Objective

The overall goal of this project is to transform **raw customer shopping data into actionable business insights** that can help the company make informed decisions regarding **sales, marketing, product strategy, customer engagement, and customer retention**.